Muhammad Jamal Nasir

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Career Profile

AI Engineer with hands-on experience in developing and deploying intelligent systems using Machine Learning, Deep Learning, NLP, Generative AI, Computer Vision and Vision-Language Models (VLMs). Skilled in building AI Agents, RAG pipelines, and automation solutions with LangChain and modern AI frameworks. Passionate about solving complex problems and contributing to cutting-edge AI innovations in a collaborative, research-driven environment.

Education

Bachelors in Artificial Intelligence
University of Management & Technology, Lahore

Professional Experience

AI Engineer — Gicoh (January 2025 – Present)

Projects

- **RAG-Based Question Answering Chatbot**
  Built a terminal-based Q&A chatbot using Retrieval-Augmented Generation (RAG) with the open-source Llama 3.2 model via Ollama. Designed for accurate, context-aware answers using local document retrieval, optimized for CPU-based deployment.
- **Route Planning AI Agent**
  Developed an intelligent agent for path optimization and logistics efficiency, leveraging AI planning and LLM-based reasoning.
- **Data Extractor Agent**
  Built an automated agent to parse, clean, and structure unstructured data from diverse sources for downstream AI tasks.
- **Real-Time Uniform Detection & Compliance System**
  Built a real-time uniform compliance system using InsightFace, YOLO, and DINOv2 for face recognition, shirt detection, and similarity-based uniform verification. Integrated FastAPI backend, SQLite database, and live streaming with a multi-threaded processing pipeline.
- **Image Classification: Cheetah vs Hyena**
  Built a CNN-based model to distinguish between Cheetah and Hyena using image datasets. Achieved high accuracy by using data augmentation and hyperparameter tuning.
- **Prediction Models using ML Algorithms**
  Implemented and compared multiple machine learning algorithms for predictive analytics on structured datasets. Evaluated models using performance metrics such as accuracy, precision, recall, F1-score, MAE, MSE, and RMSE.

Skills

- **Technical Skills:** Python, Machine Learning, Deep Learning, NLP, LLMs, Generative AI, AI Agents, Computer Vison.
- **Frameworks:** TensorFlow, Keras, Scikit-learn, LangChain, LangGraph, cv2
- **Tools:** Jupyter, Google Collab, Git, Vs Code